Security and Privacy for Machine Learning

IA-1: Problem Formulation and Preliminary Implementation

Membership Inference Attack on Machine Learning Models: A Healthcare Privacy Use Case

Problem formulation, five-module mapping and preliminary implementation

Field	Details
Student	Anjali Ashtankar
Roll number	16014223012
Batch	B1
Assessment	IA-1 (20 marks) - individual research-based assessment
Dataset	Pima Indians Diabetes (public benchmark)
Code & data	https://github.com/anjaliash005/MLCS-IA
Entry point	`src/run_ia1_baseline.py` and modules in `src/` (see Section 4)

Builds on the Module 1 research-foundation document. All numbers are produced by the code in `src/`; experiments use only public data and locally trained models.

Abstract

A machine-learning model trained on patient records can reveal whether a particular person's record was in its training set. This is a **membership inference attack (MIA)**, and in healthcare, membership alone can disclose a diagnosis. This IA-1 report formulates the problem for a diabetes-risk classifier, maps it to all five course modules, prepares the Pima Indians Diabetes dataset with privacy-safe preprocessing, and implements a black-box shadow-model attack as the baseline.

On the first split, the unregularised MLP memorises its training data (train accuracy 1.00, test 0.72) and the attack identifies members with ROC-AUC **0.738**, against 0.518 for logistic regression. Prototypes of the defence mechanisms for Modules 2-5 (DP accountant, LDP mechanisms, Mondrian k-anonymity, PCA/DCA projections) were implemented and checked. IA-2 will attack each defence with the same harness and compare privacy against utility over repeated runs.

1. Problem Definition and Module Mapping

1.1 Significance

Hospitals increasingly train predictive models on patient data and expose them through prediction services. Protecting the database is not enough: the released model itself can leak information about its training records. If a model is trained on patients enrolled in a diabetes programme, learning that a person's record was in the training set can reveal that they took part in the programme. MIAs are among the most studied privacy attacks on ML (Shokri et al., 2017), and they are a practical way to *measure* how much a model leaks.

1.2 Problem statement

Given black-box access to a diabetes-risk classifier and auxiliary data from the same population, how accurately can an attacker decide whether a specific patient's record was used for training, and which privacy-preserving techniques from the course reduce this leakage at an acceptable cost in predictive utility?

1.3 Objectives

- **O1** Build a reproducible black-box MIA (shadow models + attack classifier) against a healthcare classifier.
- **O2** Measure how leakage depends on model complexity and regularisation (baseline matrix A-D).
- **O3** Map the problem to all five modules and choose one defence family from each.
- **O4** Prepare a public dataset with preprocessing that does not itself leak private information.
- **O5** Prototype and check the defence mechanisms, and plan their evaluation for IA-2.

1.4 Scope

- **In scope:** binary tabular classification, black-box access to confidence scores, attackers with auxiliary same-distribution data, defences from Modules 1-5, utility and privacy metrics.
- **Out of scope:** white-box or gradient access, attacks on real services or real people, large neural networks and unstructured data, legal compliance analysis.
- **Expected outcome:** a quantified baseline leakage, a working attack harness, working prototypes of the defence mechanisms, and an evaluation plan that compares defences on one common privacy metric.

1.5 Module-wise concept mapping

Module	Relevant syllabus topics	Relevance to this problem	Planned treatment
M1 Privacy considerations in ML	1.1 privacy in the AI era; 1.2 threats and attacks; 1.3 PPML; 1.4 PPDM overview	Defines the threat (MIA), the healthcare harm, and model choice / regularisation as the first defence	Implemented: attack + model matrix (IA-1)
M2 Differential privacy for ML	2.1 mechanisms and properties; 2.2 DP in ML; 2.3 DP supervised; 2.4 DP unsupervised	DP bounds any single record's influence on the model, which directly limits membership inference	Accountant prototyped (IA-1); DP-GD evaluated against the attack (IA-2); DP clustering as a demonstration
M3 Local DP for ML	3.1 LDP mechanisms; 3.2 advanced LDP; 3.3 Piecewise mechanism; 3.4 LDP Naive Bayes	Patients perturb their own records before the hospital sees them, so the collector need not be trusted	Mechanisms implemented and checked against theory (IA-1); LDP-trained models attacked (IA-2)

Module	Relevant syllabus topics	Relevance to this problem	Planned treatment
M4 Privacy-preserving data mining	4.1 privacy in data mining; 4.2 modifying input; 4.3 publishing data; 4.4 beyond k-anonymity; 4.5 modifying output	Anonymising the training table and restricting API outputs are common real-world responses to leakage	Mondrian k-anonymity prototyped (IA-1); input noise, l-diversity and output restriction attacked (IA-2); 4.1 analytical
M5 Compressive privacy for ML	5.1 compressive privacy; 5.2 mechanisms; 5.3 CP for ML; 5.4 utility-task accuracy	Releasing low-dimensional projections instead of raw records can keep the diagnosis signal while hiding other attributes	PCA/DCA prototyped with utility vs k (IA-1); privacy task and MIA evaluated (IA-2)

2. Conceptual Understanding

2.1 Concept notes

Concept	Principle	Assumptions	Advantages	Limitations
Membership inference (M1)	An attacker observes the model's output M(x) for a record x and decides whether x was in the training set D. Shadow models trained by the attacker on similar data produce outputs with known membership, which are used to train an attack classifier.	Attacker has same-distribution data and knows the training recipe; black-box access to scores.	Black-box, realistic, quantitative (AUC, advantage).	Average-case metric; a weak attack can underestimate risk; needs balanced evaluation.
Overfitting and regularisation (M1)	Models that fit training data much better than unseen data give members higher confidence, so the train-test gap predicts leakage (Yeom et al., 2018). L2 penalties, early stopping and simpler models reduce memorisation.	Leakage is driven mainly by memorisation.	Cheap, often improves accuracy too.	No formal guarantee; strong attackers may still succeed.
Differential privacy (M2)	A mechanism is (ε, δ)-DP if adding or removing one record changes the output distribution by at most a factor e^ε (plus δ). DP-SGD/DP-GD clips each record's gradient and adds Gaussian noise; composition tracks the total ε over training steps.	Trusted curator; neighbouring datasets differ by one record.	Provable bound on any attacker's advantage: TPR - FPR <= (e^ε - 1 + 2δ)/(e^ε + 1).	Utility loss at small ε; ε is hard to interpret; hyperparameter tuning costs privacy.
Local differential privacy (M3)	Each user randomises their own data before sending it (Laplace, Duchi, Piecewise mechanisms; randomised response; unary encoding for frequencies). Classifiers such as Naive Bayes can be built from the perturbed reports.	Untrusted collector; many users.	No trust in the hospital needed; protects data at collection.	Much noisier than central DP; needs large user numbers.
k-anonymity and beyond (M4)	A released table is k-anonymous if every combination of quasi-identifiers (age, pregnancies, BMI) is shared by at least k records. l-diversity also requires varied sensitive values in each group. Other PPDM options perturb inputs or restrict outputs.	Attacker links records through quasi-identifiers.	Intuitive and widely used for data publishing.	No formal guarantee; homogeneity and background-knowledge attacks; information loss.
Compressive privacy (M5)	Clients release $z = W^T x$ with k << d. PCA keeps maximum variance; DCA (Discriminant Component Analysis) keeps the directions that separate the utility label, so the utility task stays accurate while other information (privacy task, reconstruction) is lost.	Projection is public; utility label known when W is learned.	Reduces what is released; can target a specific privacy attribute.	No formal guarantee; protects released data, not necessarily the trained model.

2.2 How the concepts address the problem

The attack from Module 1 is used as a **common measuring instrument**. Every other module offers a different place to intervene: during training (central DP), at data collection (LDP), on the released data or API (PPDM), or on the shared representation (compressive privacy). Measuring each one with the same attack and the same utility metrics makes them directly comparable, and shows which defences give formal guarantees and which only reduce empirical leakage.

2.3 Treatment of each concept

Treatment	Concepts
Implemented (IA-1)	Shadow-model MIA; model matrix A-D; public-bounds preprocessing; zCDP accountant; Laplace / Duchi / Piecewise / Hybrid mechanisms; Mondrian k-anonymity; PCA and DCA projections
Demonstrated (IA-1)	LDP mechanism variance vs theory; ε vs training steps (composition); k-anonymity information loss vs k; projection utility vs k
Experimentally evaluated (IA-2)	Each defence attacked by the MIA over repeated splits: DP-GD, LDP training and LDP Naive Bayes, k-anonymity / l-diversity, input noise, output restriction, PCA / DCA / DUCA
Discussed analytically	Privacy complications in the AI era (1.1); PPDM overview (1.4, 4.1); knowledge distillation as an alternative defence

2.4 Methodology and workflow

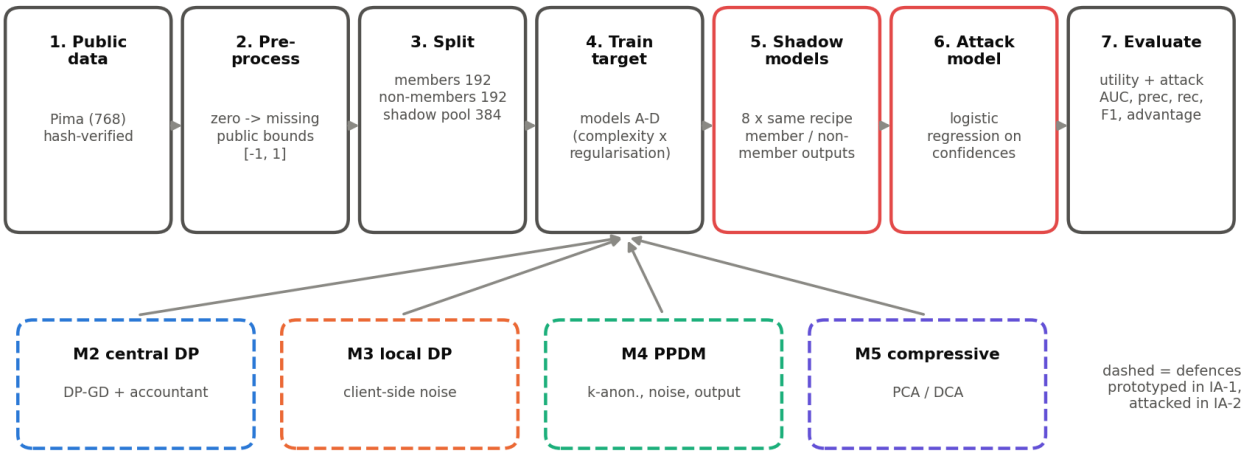


Figure 1. Proposed workflow. Solid boxes form the IA-1 baseline pipeline; dashed boxes are defences prototyped in IA-1 and attacked in IA-2.

2.5 Justification for concept selection

- **Shadow-model MIA** is the reference black-box attack and matches the IA-1 threat model (prediction API returning confidence scores).
- **DP-GD** is the standard formal defence for training and gives a bound that can be compared with the measured attack success.
- **Piecewise mechanism and LDP Naive Bayes** are the LDP methods named in the syllabus (3.3, 3.4) that suit numeric and categorical clinical data.
- **Mondrian k-anonymity** suits numeric quasi-identifiers such as age and BMI; l-diversity covers 4.4.
- **DCA** is the compressive-privacy method designed for a known utility label (diabetes), with PCA as the unsupervised comparison.

3. Dataset Identification and Preliminary Preparation

3.1 Dataset description and suitability

The **Pima Indians Diabetes** dataset contains 768 records of female patients aged 21 or older of Pima heritage, with eight clinical measurements and a binary outcome (diabetes onset within five years). It is suitable because (i) it is a real healthcare prediction task matching the use case, (ii) it is public and widely used, so no private data is put at risk, (iii) it is small enough that models can overfit, which makes membership leakage measurable, and (iv) its age, pregnancies and BMI columns act as quasi-identifiers for the k-anonymity work.

Item	Details
Source	jbrownlee/Datasets GitHub mirror of the UCI ML Repository file (raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv)
Original publisher	National Institute of Diabetes and Digestive and Kidney Diseases; Smith et al. (1988); donated to UCI in 1990
Collection method	Clinical examination: 2-hour oral glucose tolerance test, diastolic blood pressure, triceps skinfold, 2-hour serum insulin, BMI, diabetes pedigree function, age
Access date	2026-10-23 (UTC), HTTPS download
Licence / usage terms	Not stated by the mirror (no licence file). Kaggle's 'uciml' redistribution lists CC0. Treated as unknown; used only for non-commercial academic benchmarking
Version / integrity	Mirror commit 357ed4a (2018); SHA-256 6bfe5d0f379d17a0e0819b996407e3c09bf80febd4287f2ed212190dfff154af
Size	768 rows x 9 columns; outcome 0/1 = 500/268
Limitations	Small; single population and sex; zeros used for missing values; ethical concerns about continued benchmark use of data from an Indigenous community

3.2 Feature profile

Feature	Unit	Public bound	Missing (zeros)	Median (observed)
Pregnancies	count	0-17	0	3
Glucose	mg/dL (2-h OGTT)	40-200	5	117
BloodPressure	mm Hg (diastolic)	20-125	35	72
SkinThickness	mm (triceps)	5-100	227	29
Insulin	µU/mL (2-h)	10-850	374	125
BMI	kg/m²	15-68	11	32.3
DiabetesPedigreeFunction	score	0.05-2.5	0	0.3725
Age	years	21-81	0	29

3.3 Preliminary preparation log

Step	Operation	Result
1	Download and record SHA-256; verify the hash on every load	Pipeline stops if the file changes
2	Check shape and labels	768 x 9, no duplicate header, outcome 0/1 = 500/268
3	Recode physiologically impossible zeros as missing (Glucose, BloodPressure, SkinThickness, Insulin, BMI)	Missing: Glucose 5, BP 35, Skin 227, Insulin 374, BMI 11
4	Clip to fixed public clinical bounds and scale to [-1, 1]	Data-independent scaling: no statistics of private records are used, so later DP/LDP guarantees hold
5	Impute missing values with the centre of the public range; add missing indicators for SkinThickness and Insulin	10 model features; no rows dropped
6	Stratified split: 25% target members, 25% non-members (test set), 50% attacker shadow pool	192 / 192 / 384 records, positive rate 0.349 in each part

Why not standard scaling? Fitting a scaler or median imputer on the training records would use private statistics. Any DP or LDP guarantee would then have to account for them. Fixing the bounds from clinical plausibility avoids this at negligible cost.

4. Initial Implementation

4.1 Tools and implementation choices

Python 3.9 with NumPy, pandas, scikit-learn (models, metrics), SciPy and Matplotlib, pinned in `requirements.lock.txt`. Each training procedure is written as a *recipe* function `recipe(X, y, rng) -> model`. The attacker trains its shadow models with exactly the same recipe as the target, which is the strongest realistic assumption (Kerckhoffs' principle) and lets any future defence be dropped in unchanged.

Component	File	Inputs -> outputs
Data loading and preprocessing	`src/common.py`	raw CSV -> scaled feature matrix X, labels y, splits
Target models A-D	`src/models.py`	(X, y) -> fitted classifier with predict_proba
Shadow-model attack	`src/attacks.py`	recipe, data, split -> attack scores and metrics
Baseline experiment	`src/run_ia1_baseline.py`	-> `results/ia1/baseline_seed0.csv`, `baseline_log.txt`
Mechanism prototypes	`src/m2_dp.py`, `src/m3_ldp.py`, `src/m4_ppdm.py`, `src/m5_compressive.py`	accountant, LDP mechanisms, Mondrian partitioner, PCA/DCA projections

Attack procedure

- Train the target on the 192 member records.
- Train 8 shadow models, each on 192 records from the attacker's pool, with 192 other pool records as its non-members.
- For every shadow record, compute features from the model output: true-class probability, loss, entropy, maximum probability, correctness, logit and label. Label each record member (1) or non-member (0).
- Fit a standardised logistic-regression attack classifier on these labelled shadow outputs.
- Score the target's 192 members and 192 non-members and compute the attack metrics.

4.2 Baseline results (single split, seed 0)

Table 1. *Baseline utility and membership-inference results on Pima (seed 0).*

Model	Train acc	Test acc	Test F1	Gap	Attack AUC	Attack acc	Prec	Rec	F1	Adv
A: LogReg (L2)	0.766	0.760	0.589	0.005	0.518	0.484	0.487	0.583	0.531	-0.031
B: MLP-32	0.766	0.771	0.627	-0.005	0.525	0.513	0.510	0.667	0.578	0.026
C: MLP-128x2 (baseline)	1.000	0.719	0.585	0.281	0.738	0.669	0.607	0.958	0.743	0.339
D: MLP-128x2 + L2 + early stop	0.661	0.656	0.029	0.005	0.506	0.487	0.478	0.281	0.354	-0.026

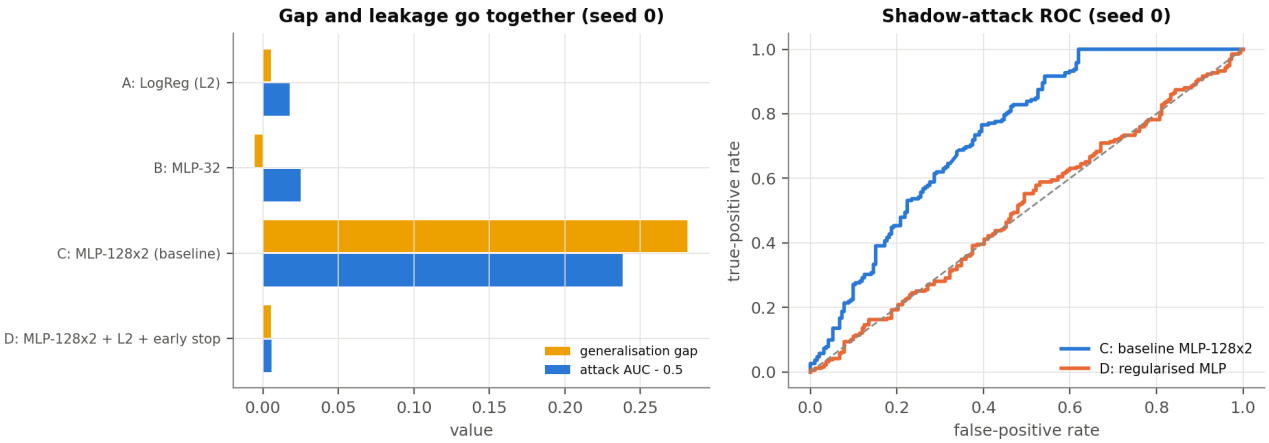


Figure 2. Left: generalisation gap and attack AUC above chance for models A-D. Right: ROC curve of the shadow attack against the baseline and the regularised MLP.

4.3 Prototype checks of the defence mechanisms

Before attacking the defences in IA-2, each mechanism was checked in isolation (Fig. 3). **M2:** the zCDP accountant reproduces sequential composition (ε grows with the number of noisy steps). For 200 steps, ε = 4 needs noise multiplier σ = 18.3, and the advantage bound at ε = 4 is 0.964. **M3:** the empirical variance of the Laplace, Duchi and

Piecewise mechanisms matches the theoretical formulas, which confirms the implementations. **M4:** Mondrian partitioning of the full table shows information loss rising with k while the share of records in diagnosis-homogeneous groups falls. **M5:** a single DCA component already matches the accuracy of the raw features, while PCA needs several components.

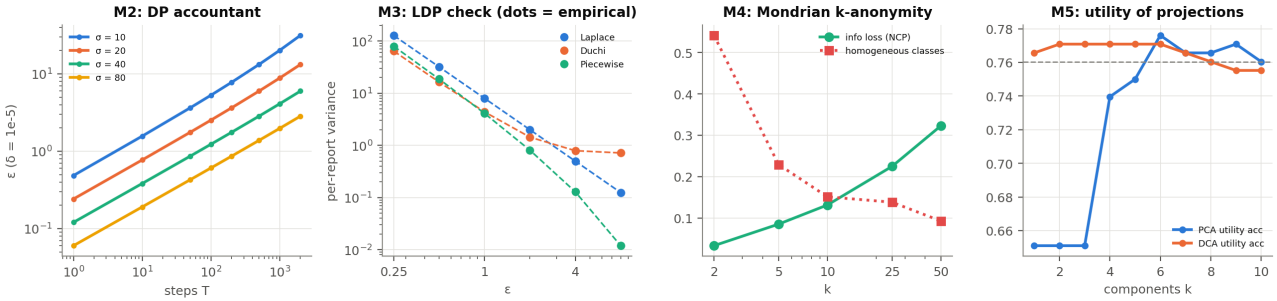


Figure 3. Mechanism-level checks: DP accounting, LDP mechanism variance (dashed = theory), k-anonymity trade-off, and utility of PCA/DCA projections (seed 0).

4.4 Execution instructions

```
git clone https://github.com/anjaliash005/MLCS-IA.git && cd MLCS-IA
python3 -m venv .venv && .venv/bin/pip install -r requirements.lock.txt
.venv/bin/python src/run_ia1_baseline.py
```

Output log

```
=== IA-1 baseline: membership inference on Pima (seed 0) ===
raw file sha256 : 6bfe5d0f379d17a0e0819b996407e3c09bf80febd4287f2ed212190dfff154af
records         : 768  (Outcome 0/1 = 500/268)
zero -> missing : Glucose=5, BloodPressure=35, SkinThickness=227, Insulin=374, BMI=11
features        : 10 (8 scaled to [-1,1] with public bounds + 2 missing indicators)
public bounds   : (see Section 3.2)
split           : target members 192, non-members 192, shadow pool 384
positive rate   : members 0.349, non-members 0.349, pool 0.349
attack          : 8 shadow models (same recipe as target) -> LR attack classifier

[A: LogReg (L2)]    0.1s
   utility : train acc 0.766 | test acc 0.760 | test F1 0.589 | test AUC 0.842 | gap 0.005
   attack  : AUC 0.518 | acc 0.484 | prec 0.487 | rec 0.583 | F1 0.531 | adv -0.031
[B: MLP-32]    1.5s
   utility : train acc 0.766 | test acc 0.771 | test F1 0.627 | test AUC 0.822 | gap -0.005
   attack  : AUC 0.525 | acc 0.513 | prec 0.510 | rec 0.667 | F1 0.578 | adv 0.026
[C: MLP-128x2 (baseline)]    5.7s
   utility : train acc 1.000 | test acc 0.719 | test F1 0.585 | test AUC 0.795 | gap 0.281
   attack  : AUC 0.738 | acc 0.669 | prec 0.607 | rec 0.958 | F1 0.743 | adv 0.339
[D: MLP-128x2 + L2 + early stop]    0.5s
   utility : train acc 0.661 | test acc 0.656 | test F1 0.029 | test AUC 0.726 | gap 0.005
   attack  : AUC 0.506 | acc 0.487 | prec 0.478 | rec 0.281 | F1 0.354 | adv -0.026

saved results/ia1/baseline_seed0.csv
```

5. Preliminary Evaluation and Documentation

5.1 Initial observations

- **Leakage is real and measurable.** The unregularised MLP (C) fits its training set perfectly (train accuracy 1.00) and the attack reaches AUC 0.738, recall 0.96 and advantage 0.34.

- **Leakage tracks overfitting.** Models with small train-test gaps (A, B, D) give attack AUCs between 0.51 and 0.52, close to random guessing.
- **Higher capacity did not buy accuracy.** Logistic regression (A) has test accuracy 0.760 vs 0.719 for model C, so on this small dataset the leaky model is also the less useful one.
- **Regularisation needs care.** On this split the heavily regularised model D collapses towards the majority class (test F1 0.029): it leaks nothing, but it is barely useful. One split is not enough to judge a defence, which motivates repeated runs and tuning in IA-2.

5.2 Evaluation measures

Aspect	Measures
Utility	Test accuracy, F1-score, ROC-AUC on held-out non-members; generalisation gap
Privacy (attack)	Attack ROC-AUC, accuracy, precision, recall, F1, membership advantage (TPR - FPR)
Formal privacy	ε (and δ) for DP / LDP configurations; comparison of measured advantage with the DP bound
Data release	Information loss (NCP), homogeneity exposure, l-diversity (M4); attribute-inference accuracy and reconstruction error (M5)
Statistics	Mean ± standard deviation over repeated random splits; paired tests against the baseline

5.3 Known limitations

- Single split: results can vary noticeably between splits on 768 records.
- Only one attacker (shadow model + logistic regression); stronger attacks may find more leakage.
- One small dataset from one population; external validity is limited.
- Defence mechanisms have only been checked in isolation, not yet attacked.

5.4 Remaining work for IA-2

Area	Planned work
Statistics	Repeat over 10 seeds; report mean ± SD; paired Wilcoxon tests against the baseline
Attack	Add a loss-threshold attack and rounded / label-only prediction APIs
M2	Train DP-GD logistic regression and MLP at ε = 0.5-8; compare measured advantage with the DP bound; DP k-means demo
M3	Train on Piecewise-perturbed records; LDP Naive Bayes; study how many users LDP needs
M4	Attack models trained on k-anonymous data; add l-diversity, input noise and output restriction
M5	Measure attribute inference, reconstruction and MIA for PCA, DCA and privacy-aware DUCA
Generalisation	Replicate on a second public medical dataset (Breast Cancer Wisconsin)
Documentation	Final technical report and research-manuscript draft

5.5 Individual-contribution statement

This IA-1 is my individual work (Anjali Ashtankar, 16014223012, B1). I contributed:

- Problem formulation, objectives, scope and the five-module concept mapping.
- Dataset selection, provenance record and the privacy-safe preprocessing design (zeros as missing, public clinical bounds, missing indicators).
- Implementation of the shadow-model attack harness, the model matrix and the baseline experiment.
- Prototype implementations and checks of the DP accountant, LDP mechanisms, Mondrian k-anonymity and PCA/DCA projections.
- Analysis of the preliminary results and the evaluation plan for IA-2.

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